

Moment of a Force and Equilibrium

Two lessons in one sheet: the turning effect of a force, and the equilibrium of a body under several forces.

Mr. Gemy
PHYSICS

Grade 11 · Egyptian Baccalaureate

30 questions

Step-by-step solutions

$g = 9.8 \text{ m/s}^2$ where stated

The toolkit for these two lessons

$$M = F \times d$$

force $\perp$ to the arm

$$M = F L \sin \theta$$

force inclined at θ to the arm

$$\text{unit : N}\cdot\text{m}$$

moment is a vector quantity

$$\Sigma F = 0$$

condition of translational equilibrium

$$\Sigma F_x = 0, \Sigma F_y = 0$$

applied on both axes separately

$$T \sin \theta = W$$

the vertical component carries the weight

Two ideas, one sheet. A moment measures how strongly a force **turns** a body about a point, and it needs the **perpendicular** distance. Equilibrium means the forces cancel on **each axis separately**, which is why an inclined force must always be resolved first.

LESSON 1–4 · CLASSWORK 1

A The moment of a force

1 The expression for the moment M of a force when the force is **perpendicular** to the arm is:

MCQ

A $M = F / d$

B $M = F d$

C $M = F + d$

D $M = F - d$

2 The moment of a force is a measure of:

MCQ

A the magnitude of the force only

B the turning effect of the force about a point

C the speed of the body

D the energy stored in the body

3 The SI unit of the moment of a force is:

MCQ

A newton (N)

B joule (J)

C newton · metre (N·m)

D metre / second (m/s)

4 The law of the lever was formulated in antiquity by:

MCQ

A Newton

B Archimedes

C Galileo

D Kepler

5 A force of 5.0 N acts **perpendicular** to an arm of length 2.0 m measured from the point of support. Calculate the moment of the force.

PROBLEM

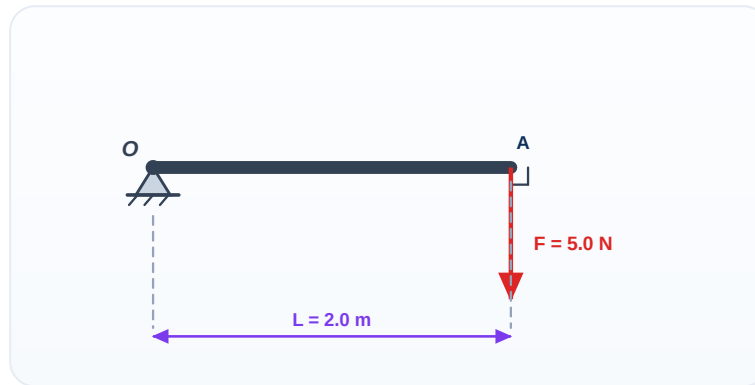


FIGURE 1

LESSON 1-4 · HOME ASSIGNMENT 1

B Using the moment relation

6 If the perpendicular distance is doubled while the force stays constant, then the moment of the force:

MCQ

A is halved

B is doubled

C stays constant

D is reduced to a quarter

7 A door opens most easily when it is pushed at:

MCQ



FIGURE 2

A the hinge itself

B the handle, far from the hinge

C the middle of the door only

D any point — it makes no difference

- 8 A moment of $12 \text{ N}\cdot\text{m}$ is produced by a force of 4.0 N acting perpendicular to its arm. The length of the perpendicular arm equals:

MCQ

- A 3.0 m B 8.0 m
C 48 m D 33 m

- 9 A force of 3.0 N acts perpendicular at the end of a horizontal rod, at a distance of 20 m from the point O. Calculate the moment of the force about O.

PROBLEM

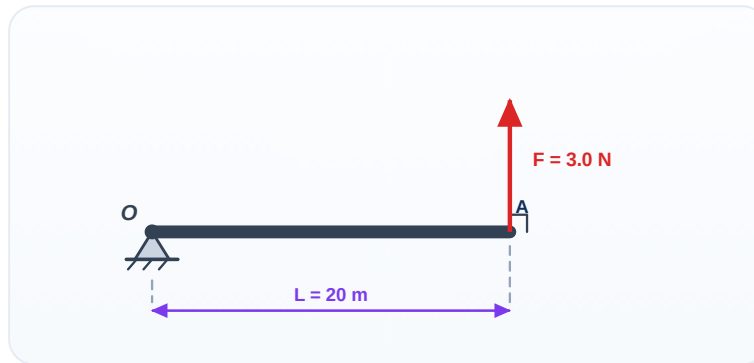


FIGURE 3

LESSON 1-5 · CLASSWORK 2

C The equilibrium of a body

- 10 A body is in a state of equilibrium when:

MCQ

- A its velocity is always zero B it stays at rest or moves with a constant velocity
C its velocity changes continuously D its mass vanishes

- 11 The condition of **translational** equilibrium of a body is written mathematically as:

MCQ

- A $\Sigma F = mg$ B $\Sigma F = 0$
C $\Sigma F = ma$, where $a \neq 0$ D $\Sigma F = F_1 - F_2$ only

- 12 To deal with a force acting at an angle, it is resolved into two perpendicular components, and the two equilibrium equations are then applied to:

MCQ

- A one axis only B both axes x and y : $\Sigma F_x = 0$ and $\Sigma F_y = 0$
C the angle θ only D the mass only

- 13 The figure shows a wire fixed between two poles, carrying a weight at its middle. **Explain how the angle θ at the two fixing points changes** if the wire is allowed to sag more at its middle, and what is the expected effect on the tension T .

EXPLAIN

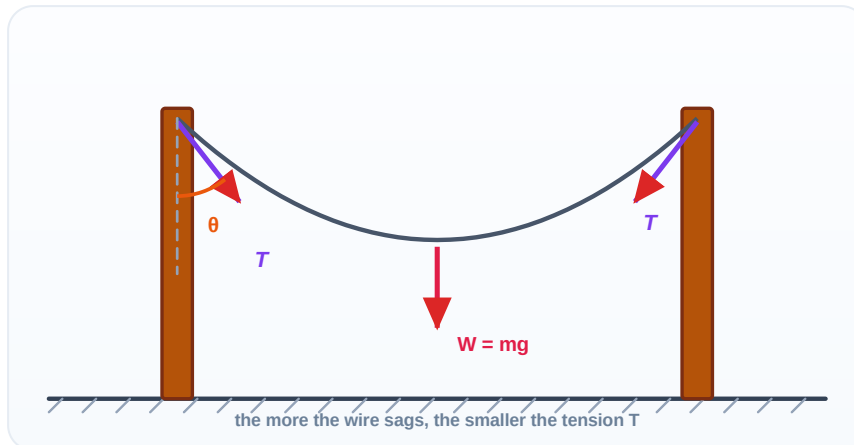


FIGURE 4

Hanging bodies and the tension in cables

- 14 A lamp hangs from a cable fixed to the ceiling, together with a light horizontal rope fixed to a wall. The cable makes an angle of 25° with the vertical. The **vertical** component of the tension in the cable must be equal to:

MCQ

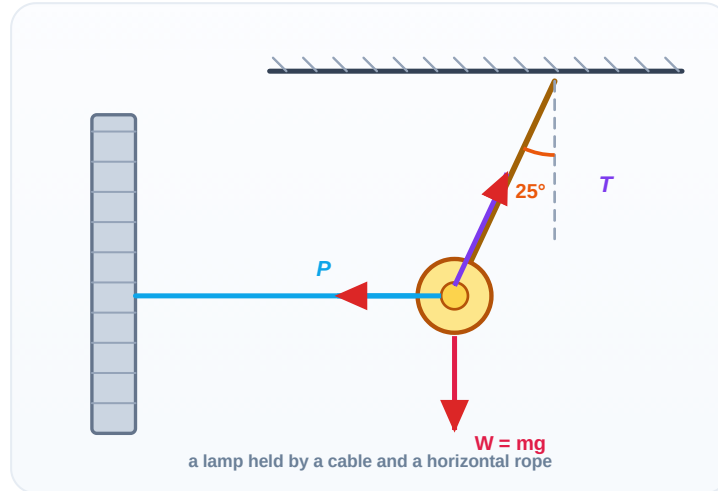


FIGURE 5

- A zero
- B the weight of the lamp
- C the tension in the horizontal rope
- D half the weight of the lamp

- 15 A lamp of mass 0.6 kg hangs from two identical cables, each making an angle of 30° with the horizontal. If $g = 9.8 \text{ m/s}^2$, then the weight of the lamp is approximately:

MCQ

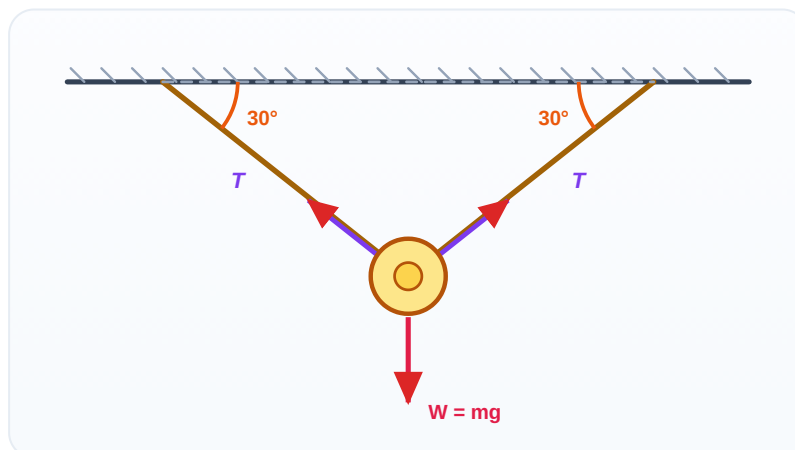


FIGURE 6

- A 0.6 N
- B 58.8 N
- C 4.29 N
- D 5.88 N

16 If the angle that each cable makes with the horizontal is increased (so that the two cables become closer to the vertical) while the load stays constant, then the tension in each cable:

MCQ

A increases

B decreases

C stays constant

D becomes zero

17 The figure shows the tension in a cable resolved into two components. **Explain which of the two components** (the horizontal one or the vertical one) is responsible for balancing the weight of the hanging body, and **why the other component is left out** of the vertical equilibrium equation.

EXPLAIN

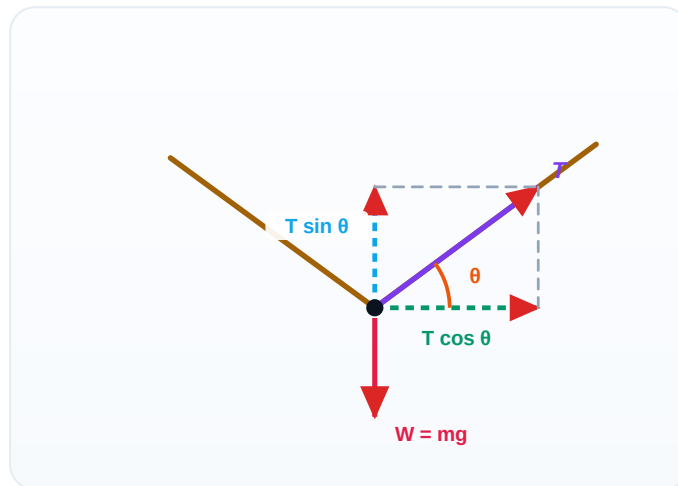


FIGURE 7

WEEKLY ASSESSMENT 3

E

Groups A, B and C

Group A

18 If the force is doubled while the perpendicular distance stays constant, then the moment of the force:

MCQ

A is doubled

B is halved

C stays constant

D cannot be determined

Mr. Gemy

Mr. Gemy

19 A wheel-nut spanner loosens the nut more easily when it is:

MCQ

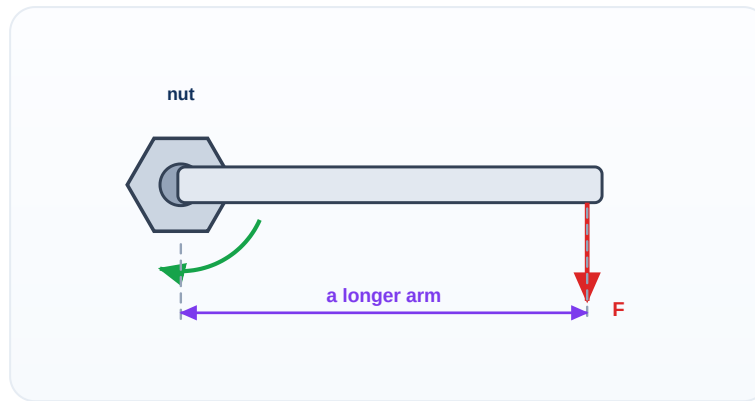


FIGURE 8

A shorter

B longer

C without an arm

D its length makes no difference

20 **Compare** the moment of a force acting perpendicular to its arm ($\theta = 90^\circ$) with the moment of the **same** force acting at an angle smaller than 90° on the **same** arm, in terms of the magnitude of the moment and the relation used in each case.

COMPARE

21 The figure shows a force F acting at the end of a lever. **How can the moment of this force about the other end of the lever be increased in three different ways**, without changing the magnitude of the force itself?

EXPLAIN



FIGURE 9

Group B

- 22 A force of 4.0 N acts at the end of an arm of length 0.30 m, making an angle of 45° with the arm. The moment of the force about the point of support is approximately: ($\sin 45^\circ \approx 0.71$)

MCQ

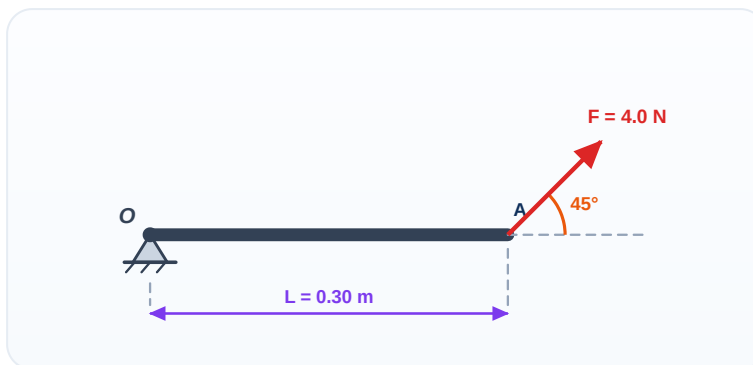


FIGURE 10

- A 0.85 N·m
B 1.2 N·m
C 1.7 N·m
D 6.0 N·m

- 23 A ruler is balanced on a point of support. One coin placed at a distance of 15 cm on the right-hand side balances n identical coins placed at a distance of 5 cm on the left-hand side. The number of coins n equals:

MCQ

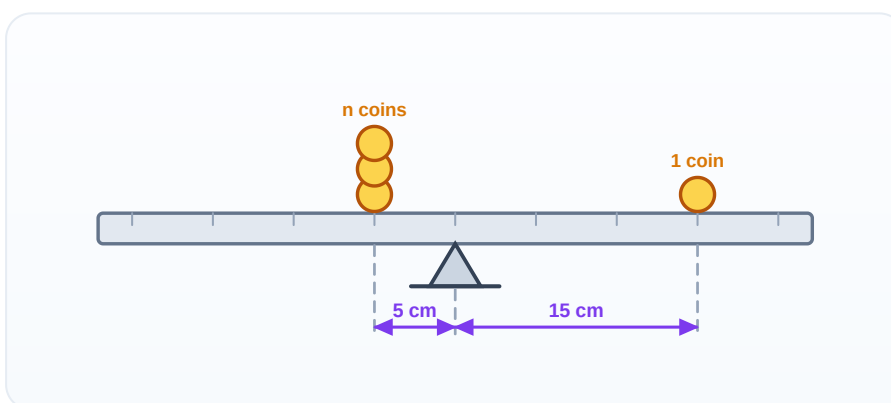


FIGURE 11

- A 1
B 2
C 3
D 5

- 24 **Compare** the effect of pushing a door at the handle far from the hinge with the effect of pushing it close to the hinge, in terms of the length of the moment arm and the magnitude of the resulting moment, with the same applied force.

COMPARE

- 25 Using the figure, explain — **without any numerical calculation** — why the magnitude of the moment of the same force differs according to the point (A or B) about which the moment is calculated, although the force has the same magnitude and the same direction in both cases.

EXPLAIN

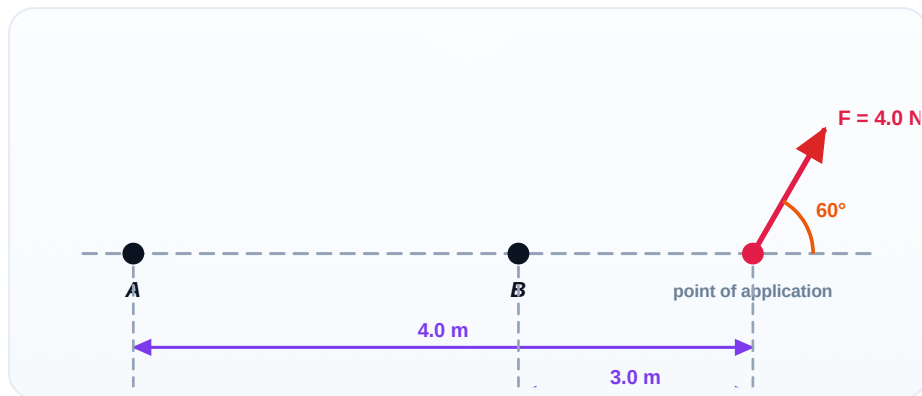


FIGURE 12

Group C

- 26 When a spanner with a **longer** arm is used to produce the same required moment, the force needed is:

MCQ

- ☐ A larger
- ☐ B smaller
- ☐ C exactly the same
- ☐ D zero

- 27 A force of 6.0 N acts at the end of a lever of length 0.50 m , at an angle of 30° to the lever. The moment of the force about the other end equals approximately: ($\sin 30^\circ = 0.5$)

MCQ

- ☐ A $1.5 \text{ N}\cdot\text{m}$
- ☐ B $0.3 \text{ N}\cdot\text{m}$
- ☐ C $0.5 \text{ N}\cdot\text{m}$
- ☐ D $0.9 \text{ N}\cdot\text{m}$

- 28 A force $F = 4.0 \text{ N}$ acts at an angle of 60° at a point which is 4.0 m from A and 3.0 m from B, where A and B lie on the same straight line. ($\sin 60^\circ \approx 0.87$)

MCQ

- 1) The moment of the force about A equals approximately: a) $13.9 \text{ N}\cdot\text{m}$ anticlockwise b) $10.4 \text{ N}\cdot\text{m}$ clockwise
c) $13.9 \text{ N}\cdot\text{m}$ clockwise d) $4.0 \text{ N}\cdot\text{m}$ anticlockwise
- 2) The moment of the same force about B equals approximately: a) $13.9 \text{ N}\cdot\text{m}$ anticlockwise b) $10.4 \text{ N}\cdot\text{m}$ clockwise
c) $10.4 \text{ N}\cdot\text{m}$ anticlockwise d) zero

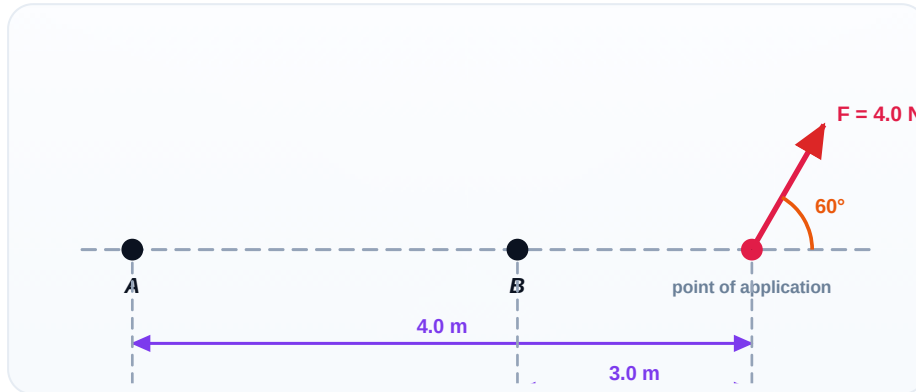


FIGURE 13

- 29 **Compare** a positive moment with a negative moment, in terms of the sense of rotation that each one represents.

COMPARE

- 30 Using the figure, explain — **without any numerical calculation** — how the moment of the force about O changes if the point of application of F slides along the rod towards O, while the direction and the magnitude of the force stay the same.

EXPLAIN

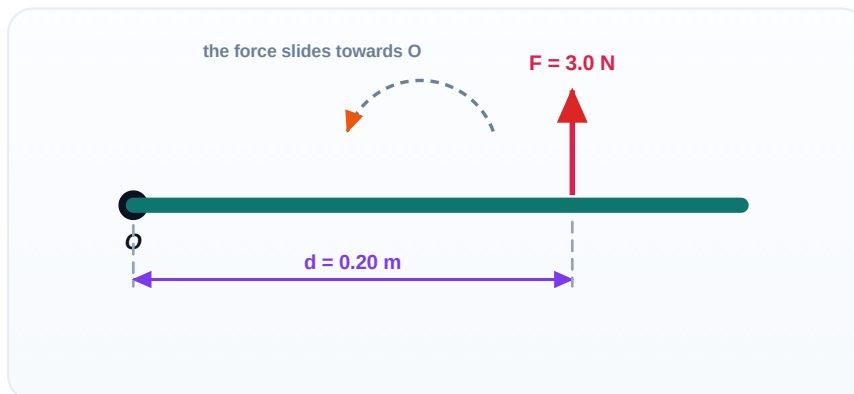


FIGURE 14



ANSWER KEY

Solutions — step by step

Remember the two habits that save marks here: always use the **perpendicular** distance in a moment, and always resolve an inclined force before writing an equilibrium equation.

Part A — Lesson 1-4 · Classwork 1

1 the basic relation

moment = force $\times$ perpendicular distance $M = F d$

B) $M = F d$

2 what a moment measures

a force can move a body (translation) or turn it about a point (rotation);
the moment measures the second effect

B) the turning effect of the force about a point

3 the unit

$M = F \times d = \text{newton} \times \text{metre} = \text{N} \cdot \text{m}$

It is never written as a joule, even though a joule is also a newton $\times$ a metre; work and moment are different physical quantities.

C) newton $\cdot$ metre

4 the law of the lever

formulated by Archimedes in antiquity

B) Archimedes

5 a perpendicular force

the force is perpendicular to the arm, so the arm itself is the moment arm :

$M = F \times L = 5.0 \times 2.0 = 10 \text{ N} \cdot \text{m}$

$M = 10 \text{ N} \cdot \text{m}$

Part B — Lesson 1-4 · Home assignment 1

6 doubling the arm

$M = F d$, with F constant $\rightarrow M$ is proportional to d

B) it is doubled

7 where to push a door

pushing at the handle gives the largest perpendicular distance from the hinge, so it gives the largest moment

B) the handle, far from the hinge

8 finding the arm

$$M = F d \rightarrow d = M / F = 12 / 4.0 = 3.0 \text{ m}$$

A) 3.0 m

9 moment about O

$$M = F \times d = 3.0 \times 20 = 60 \text{ N}\cdot\text{m}$$

M = 60 N·m

Part C — Lesson 1-5 · Classwork 2

10 the meaning of equilibrium

equilibrium means zero resultant force, so the velocity does not change : the body is either at rest or moving uniformly in a straight line

B) it stays at rest or moves with a constant velocity

11 the mathematical condition

$$\Sigma F = 0 \quad (\text{the vector sum of all the forces is zero})$$

B) $\Sigma F = 0$

12 how the condition is applied

a vector equation is equivalent to one equation for each axis :

$$\Sigma F(x) = 0 \quad \text{and} \quad \Sigma F(y) = 0$$

B) on both axes x and y

13 the sagging wire

at the lowest point, the two tensions carry the weight between them :

$$2 T \sin \alpha = W \quad (\alpha = \text{the angle between the wire and the horizontal})$$

$$T = W / (2 \sin \alpha)$$

more sag $\rightarrow \alpha$ becomes larger $\rightarrow \sin \alpha$ becomes larger $\rightarrow T$ becomes smaller
measured from the pole (the vertical), the angle θ becomes smaller.

This is why a clothes line that is pulled almost perfectly horizontal snaps so easily: as the angle with the horizontal goes to zero, the tension needed to carry the same weight grows without limit.

a bigger sag makes θ smaller and reduces the tension T

Part D — Lesson 1-5 · Home assignment 2

14 the lamp on a cable and a rope

the lamp is in equilibrium, so for the vertical axis :

$$\Sigma F(y) = 0 \rightarrow T \cos 25^\circ - W = 0$$

the horizontal rope is horizontal, so it adds nothing to the vertical equation

B) the weight of the lamp

15 the weight of the lamp

$$W = m g = 0.6 \times 9.8 = 5.88 \text{ N}$$

The mass in kilograms is not a weight; it must be multiplied by g to give a force in newtons.

D) 5.88 N

16 raising the angle of the cables

$$2 T \sin \alpha = W \rightarrow T = W / (2 \sin \alpha)$$

α increases $\rightarrow \sin \alpha$ increases $\rightarrow T$ decreases

B) decreases

17 which component carries the weight

$$\text{vertical axis : } \Sigma F(y) = 0 \rightarrow T \sin \theta = W$$

$$\text{horizontal axis : } \Sigma F(x) = 0 \rightarrow T \cos \theta \text{ is balanced by another horizontal force}$$

The horizontal component has no projection on the vertical axis at all, so it simply does not appear in the vertical equation — it is not "neglected", it is balanced separately by the other horizontal force.

the vertical component $T \sin \theta$ balances the weight

Part E — Weekly assessment 3 · Group A

18 doubling the force

$$M = F d, \text{ with } d \text{ constant} \rightarrow M \text{ is proportional to } F$$

A) it is doubled

19 the length of the spanner

$$M = F d : \text{ a longer spanner gives a larger } d, \text{ so the same force produces a larger moment}$$

B) longer

20 perpendicular force against an inclined force

$\theta = 90^\circ$: $M = F L$ ($\sin 90^\circ = 1$) $\rightarrow$ the largest possible moment

$\theta < 90^\circ$: $M = F L \sin \theta$ ($\sin \theta < 1$) $\rightarrow$ a smaller moment

Only the component of the force perpendicular to the arm turns the body; the component along the arm pulls or pushes it and has no moment.

the perpendicular force always gives the larger moment

21 three ways to increase the moment

$M = F L \sin \theta$, with F fixed :

- 1) use a longer arm L
- 2) change the direction of the force so that $\theta = 90^\circ$ ($\sin \theta = 1$)
- 3) apply the force at the point farthest from the axis of rotation,
which is the same as increasing the perpendicular distance $d = L \sin \theta$

longer arm , $\theta = 90^\circ$, or a point farther from the axis

Group B

22 an inclined force on a short arm

$$M = F L \sin \theta = 4.0 \times 0.30 \times \sin 45^\circ$$

$$M = 4.0 \times 0.30 \times 0.71 = 0.85 \text{ N}\cdot\text{m}$$

The common trap here is to answer $1.2 \text{ N}\cdot\text{m}$, which is $F \times L$ without $\sin \theta$ — that value only applies when the force is perpendicular to the arm.

A) $0.85 \text{ N}\cdot\text{m}$

23 the balanced ruler

the two moments about the point of support are equal :

$$(1 \text{ coin}) \times 15 = (n \text{ coins}) \times 5$$

$$15 = 5n \rightarrow n = 3$$

C) 3 coins

24 pushing a door near and far from the hinge

far from the hinge : large moment arm $\rightarrow$ large moment $\rightarrow$ the door turns easily

close to the hinge : small moment arm $\rightarrow$ small moment $\rightarrow$ the door hardly turns

the same force gives a much larger moment when it is applied far from the hinge

25 why the moment changes with the point

$M = F \times d$, and d is the perpendicular distance from the **chosen point** to the line of action of the force.

A and B are at different distances from that line, so each gives a different d , and a point lying on the line of action itself gives $d = 0$ and $M = 0$.

The force never changes; what changes is the geometry between the force and the point you choose to turn about.

because each point has its own perpendicular distance d

Group C

26 a longer spanner for the same moment

$M = F d$, with M fixed $\rightarrow F = M / d$

d larger $\rightarrow F$ smaller

B) smaller

27 inclined force on a lever

$M = F L \sin \theta = 6.0 \times 0.50 \times \sin 30^\circ$

$M = 6.0 \times 0.50 \times 0.5 = 1.5 \text{ N}\cdot\text{m}$

A) 1.5 N·m

28 the same force about two different points

1) about A : $d(A) = 4.0 \sin 60^\circ = 4.0 \times 0.87 = 3.48 \text{ m}$

$M(A) = F \times d(A) = 4.0 \times 3.48 = 13.9 \text{ N}\cdot\text{m}$

the force turns the line anticlockwise about A

2) about B : $d(B) = 3.0 \sin 60^\circ = 2.61 \text{ m}$

$M(B) = 4.0 \times 2.61 = 10.4 \text{ N}\cdot\text{m}$, anticlockwise as well

1) a) 13.9 N·m anticlockwise · 2) c) 10.4 N·m anticlockwise

29 the sign of a moment

positive moment $\rightarrow$ anticlockwise rotation

negative moment $\rightarrow$ clockwise rotation

The sign lets moments be added like ordinary numbers: a positive and a negative moment of the same size cancel each other exactly.

the sign is only a label for the sense of rotation

30 sliding the force towards O

$M = F \times d$, with F fixed in size and direction

as the point of application slides towards O, the perpendicular distance d gets smaller, so the moment decreases in the same ratio, until $d = 0$ at O and the moment vanishes

At O the line of action of the force passes through the point itself, and a force whose line of action passes through a point can never turn the body about that point.

the moment decreases steadily and becomes zero at O